Natural Language Inference Literature Leaderboards

GraphNLI-AXb project

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Scope and protocol

This document summarizes published results for Natural Language Inference (NLI) benchmarks used in the GraphNLI-AXb project. NLI evaluates whether a hypothesis is entailed by, contradicts, or is neutral with respect to a premise. The covered resources span English, multilingual and Portuguese evaluation settings: SICK, XNLI, ASSIN, ASSIN2, InferBR and AX-b.

The tables are organized by dataset because their label spaces, splits and evaluation protocols are different. SICK, XNLI and InferBR generally use three-way classification and report accuracy. ASSIN uses the three classes entailment, paraphrase and neutral, while ASSIN2 uses binary entailment versus non-entailment. Portuguese studies commonly report accuracy and macro-F1.

Results are ranked only within a table and only when their metric and protocol are sufficiently aligned. In particular, zero-shot, translate-train, translate-test and supervised fine-tuning are reported explicitly rather than being silently combined. Values marked as development, cross-dataset transfer or a different metric are contextual references and should not be interpreted as direct leaderboard entries.

AX-b

AX-b (Broadcoverage Diagnostics) is a diagnostic NLI set derived from GLUE Diagnostics and included in SuperGLUE. It focuses on linguistic phenomena such as negation, quantifiers, conjunction, conditionals, monotonicity and temporal reasoning. SuperGLUE maps contradiction and neutral to the binary label `not_entailment`, so AX-b is evaluated with Matthews correlation coefficient (MCC) rather than three-way accuracy [1].

The official leaderboard reports MCC multiplied by 100. Thus, a leaderboard value of 72.9 corresponds to MCC = 0.729 on the usual $[-1, 1]$ scale. The accuracy table is kept separate because OpenCompass reports a different metric and protocol.

Table 1 collects the strongest publicly documented AX-b references found in the literature review. Results are deliberately not merged into a single numerical ranking when the metric or evaluation protocol changes. In particular, the 2024 PanGu-π paper reports accuracy through OpenCompass [19], whereas the official SuperGLUE diagnostic uses MCC.

MCC leaderboard

Table 1: AX-b literature leaderboard. MCC values are multiplied by 100, as in the official SuperGLUE leaderboard.

Rank	Year	System	Total parameters	AX-b MCC	Primary reference or provenance
1	2019	SuperGLUE human baseline	–	76.6	Human reference [1]
2	2021	PAI Albert	N/D	75.6	Official leaderboard submission; primary paper not located

3	2023	PaLM 540B	540B	72.9	Total size is documented in the literature [3]
4	2022	ST-MoE-32B	269B	72.3	Total size reported in the SPoT comparison table; sparse model [4, 8]
5	2021	T5 + UDG	11B	69.1	Total size reported in the SPoT comparison table [5, 8]
6	2021	ERNIE 3.0	12B	68.6	Total size reported in the SPoT comparison table [6, 8]
7	2021	Turing NLR v5	5.4B	67.8	Total size reported in the SPoT comparison table [7, 8]
8	2022	SPoT (Frozen T5 1.1)	11B	66.9	Total size reported by the authors [8]
9	2021	DeBERTa / TuringNLRv4	3.1B	66.7	Total size reported in the SPoT comparison table [9, 8]
10	2020	T5	11B	65.6	Total size reported in the SPoT comparison table [10, 8]
11	2023	RoBERTa-iCETS	N/D	61.8	Official leaderboard submission; primary paper not located
12	2022	RoBERTa-mtl-adv	N/D	58.5	Official leaderboard submission; primary paper not located
13	2021	NEZHA-Plus	N/D	58.0	Official leaderboard submission; primary paper not located
14	2019	Facebook RoBERTa	N/D	57.9	Official model variant and tuning details are not specified in the leaderboard row [11]
15	2022	RoBERTa (ensemble)	N/D	57.6	Official leaderboard submission; exact ensemble composition not documented
16	2020	Snorkel	N/D	47.6	Official leaderboard submission, version 1.9
17	2019	BERT++	N/D	38.0	Original baseline; exact parameterization is not separately reported [1]
18	2021	iPET (ALBERT), 32 examples	N/D	36.2	Primary paper; exact leaderboard checkpoint parameterization is not separately reported [12]
19	2021	FSL++ (ALBERT), 32 examples	N/D	35.9	Official leaderboard submission; primary paper not located
20	2020	BERT-mtl	N/D	29.6	Official leaderboard submission; exact backbone size not documented
21	2019	BERT-large	340M	23.0	BERT-Large size documented in the baseline setup [1, 2]
22	2020	GPT-3 few-shot	175B	21.1	Total size reported in the SPoT comparison table [13, 8]
23	2019	Most Frequent Class	–	0.0	Original baseline [1]
24	2021	ADAPET (ALBERT), few-shot	N/D	-0.4	Primary paper; exact leaderboard checkpoint parameterization is not separately reported [14]

25	2023	Vega v2	N/D	-0.4	Primary technical report; exact submitted checkpoint size is not stated in the leaderboard row [15]
26	2021	INSTALL (ALBERT), few-shot	N/D	-0.4	Official leaderboard submission; primary paper not located
27	2021	WARP (ALBERT-XXL), few-shot	223M	-0.4	Total size reported in the SPoT comparison table [8]
28	2019	CBoW	15M	-0.4	Total size reported in the SPoT comparison table [8]
29	2021	Select-step-by-step	N/D	-0.4	Official leaderboard submission; primary paper not located

Accuracy leaderboard

Table 2: AX-b results reported with accuracy rather than MCC. Values are reported by Tang et al. [17] using OpenCompass [19]. These values must not be ranked together with Table 1.

Rank	Year	System	Total parameters	AX-b accuracy	Primary reference or provenance
1	2024	PanGu-π-1.5B Pro	1.5B	67.93	Model size documented in the primary paper; OpenCompass accuracy, not MCC [17, 19]
2	2024	Qwen-1.8B	1.8B	57.79	Model size documented in the primary paper; OpenCompass accuracy, not MCC [17, 19]
3	2024	MobileLLaMA-2.7B	2.7B	55.80	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
4	2024	RWKV-5-1.5B	1.5B	54.05	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
5	2024	Open-LLaMA-3B	3B	52.72	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
6	2022	BLOOM-1B1, one-shot	1.1B	52.72	One-shot evaluation; BigScience artifact [20]
7	2024	Chinese-LLaMA2-1.3B	1.3B	47.46	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
8	2024	TinyLLaMA-1.1B	1.1B	45.47	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
9	2024	MobileLLaMA-1.4B	1.4B	45.20	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]

10	2024	Phi-1.3B	1.3B	44.20	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
11	2024	Sheared-LLaMA-1.3B	1.3B	43.57	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
11	2024	Phi-2.7B	2.7B	43.57	Model size is part of the evaluated model identifier; OpenCompass accuracy, not MCC [19]
13	2024	PanGu-π-1B	1B	43.48	Model size documented in the primary paper; OpenCompass accuracy, not MCC [17, 19]

SICK

Table 3: SICK test-set leaderboard. All systems are trained on the original SICK training split and evaluated on the original three-way SICK test set.

Rank	Year	System	Parameters	Accuracy	Reference
1	2023	RoBERTa-Large	355M	91.50	Sharma et al. [24]
2	2023	RoBERTa-Large	355M	90.5	Kalouli et al. [23]
3	2023	ALBERT-Large	17M	89.85	Sharma et al. [24]
4	2023	BERT-Large	340M	89.85	Sharma et al. [24]
5	2023	BERT-Base	110M	85.71	Sharma et al. [24]
6	2023	BERT-Large	340M	85.3	Kalouli et al. [23]
7	2023	DAM	382K	83.37	Sharma et al. [24]
8	2023	ESIM	4.3M	80.81	Sharma et al. [24]

The Sharma et al. results are an independent reimplementation study, while Kalouli et al. use SICK to analyze annotation regimes. The small numerical difference between the two RoBERTa/BERT evaluations should therefore not be interpreted as a strict leaderboard improvement.

XNLI

Table 4: XNLI average accuracy across the 15 official languages.

Rank	Year	System	Protocol	Accuracy	Reference
1	2025	Gemma 2 9B + FLARE	Translate-train	85.01	Borchert et al. [29]
2	2023	RoBERTa-Large + NLLB	Translate-test	83.8	Artetxe et al. [28]
3	2022	XLM-R-Large + PCT	Full-shot + translate-train-all	83.6	Qi et al. [27]
4	2020	XLM-R-Large	Translate-train-all	82.4	Conneau et al. [26]
5	2020	XLM-R-Large	Zero-shot transfer	79.2	Xue et al. [30]
6	2020	mBERT	Zero-shot transfer	65.4	Xue et al. [30]

The XNLI table intentionally exposes the protocol. Translate-train and translate-test use additional machine-translated inputs, whereas zero-shot transfer uses English training data and evaluates directly in the target languages. These regimes should be compared only within the same protocol.

ASSIN

Table 5: ASSIN three-class NLI leaderboard on the combined Portuguese European/Brazilian test setting. The classes are entailment, paraphrase and neutral.

Rank	Year	System	Parameters	Accuracy	Macro-F1	Reference
1	2020	ptBERT-Large fine-tuned	335M	91.0	82.0	Fialho et al. [31]
2	2020	ptBERT-Base fine-tuned	110M	90.0	81.0	Fialho et al. [31]
3	2020	mBERT-Base fine-tuned	178M	90.0	80.0	Fialho et al. [31]
4	2016	ASSIN challenge best system	N/D	83.85	71.0	Fonseca et al. [32]

ASSIN is not identical to the binary NLI formulation used by ASSIN2 or InferBR. In particular, paraphrase is a separate class here; it should not be collapsed into neutral when comparing these values.

ASSIN2

Table 6: ASSIN2 binary NLI leaderboard. The task labels are entailment and non-entailment; the official test split is used.

Rank	Year	System	Parameters	Accuracy	Macro-F1	Reference
1	2020	BERTimbau-Large fine-tuned	335M	90.0	90.0	Souza et al. [33]
2	2026	BERTomelo-Large fine-tuned	N/D	89.26	89.21	Oliveira et al. [35]
3	2020	BERTimbau-Base fine-tuned	110M	89.2	89.2	Souza et al. [33]
4	2020	mBERT fine-tuned	178M	86.8	86.8	Souza et al. [33]
5	2020	mBERT + RoBERTa-Large ensemble	N/D	88.3	88.3	Rodrigues et al. [34]

NorBERTo reports approximately 0.904 entailment F1 on ASSIN2, but this is a per-class F1 rather than macro-F1. It is therefore recorded as a recent reference, not inserted into the ranking above [36].

InferBR

Table 7: InferBR three-class NLI leaderboard. Results are from the original InferBR evaluation protocol, with BERTimbau fine-tuned on each dataset's training and validation splits.

Rank	Year	System	Parameters	Accuracy	Macro-F1	Reference
1	2024	BERTimbau fine-tuned on InferBR	N/D	90.0	90.0	Bencke et al. [37]
2	2024	BERTimbau trained on SICK-BR, tested on InferBR	N/D	60.0	58.0	Bencke et al. [37]
3	2024	BERTimbau trained on InferBR, tested on SICK-BR	N/D	64.0	63.0	Bencke et al. [37]

The cross-dataset rows are included as transfer references, not as direct InferBR leaderboard entries. They illustrate the domain shift between synthetically generated InferBR examples and SICK-BR examples.

Here "Total parameters" refers to the complete model size. "N/D" means that the primary source or official submission does not document the total parameter count for that exact entry; it is intentionally not inferred from the model name. For sparse models, total parameters include all experts. The leaderboard is a historical snapshot and mixes official submissions, human references, few-shot systems, and baselines. The 2024–2026 review found no newer peer-reviewed result with a clearly documented official AX-b MCC above 72.9; recent papers often omit AX-b or report a different metric and split.

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